Constellation

SVP-24-063 10 CFR 50.73

October 30, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 2
Renewed Facility Operating License No. DPR-30
NRC Docket No. 50-265

Subject: Licensee Event Report 265/2024-002-01 "Turbine Trip and Automatic Scram due to Digital EHC Power Supply Intermittent Failure" Supplement

Reference: Letter, D Hild to Document Control Desk – Licensee Event Report 265/2024-002-00 "Turbine Trip and Automatic Scram due to Digital EHC Power Supply Intermittent Failure," July 22, 2024 (ML24204A105)

Enclosed is Licensee Event Report 265/2024-002-01 "Turbine Trip and Automatic Scram due to Digital EHC Power Supply Intermittent Failure," for Quad Cities Nuclear Power Station, Unit 2. This letter is a supplement based on the completed causal evaluation. The updated information is denoted by revision bars located in the left-hand margin.

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(iv)(A) for an event or condition that resulted in manual or automatic actuation of the reactor protection system including a reactor scram, and for containment isolation signals affecting more than one system.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Conner Bealer at 779-231-6207.

Respectfully,

Doug Hild
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027
LICENSEE EVENT REPORT (LER) **(See Page 2 for required number of digits/characters for each block)** (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name		2. Docket Number	3. Page
Quad Cities Nuclear Power Station, Unit 2	☒ 050 ☐ 052	00265	1 OF 4

4. Title

Turbine Trip and Automatic Scram due to Digital EHC Power Supply Intermittent Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
05	23	2024	2024	- 002 -	01	10	30	2024	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1 – Power Operation	038

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Richard Swart	309-227-2810

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	TG	RJX	G080	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On May 23, 2024, at 2223 CST, Quad Cities Unit 2 had an automatic scram from 38 percent power due to a trip of the main turbine. A Digital Electro-Hydraulic Control (DEHC) erroneous Speed Difference Trip signal initiated a Turbine Trip signal. The Unit 2 Main Turbine tripped as designed and generated an automatic reactor scram. All control rods inserted and the scram was uncomplicated. Containment isolation valves actuated closed in multiple systems on valid Group II and Group III signals as a result of low reactor water level.

The cause of the DEHC trip signal was intermittent channel <R1> core power supply failure, resulting in an erroneous speed demand value being generated. The power supply and several DEHC cards were replaced with new components.

This report is being submitted per 10 CFR 50.73(a)(2)(iv)(A) for an event that resulted in manual or automatic actuation of the reactor protection system including a reactor scram, and containment isolation signals affecting more than one system.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Quad Cities Nuclear Power Station, Unit 2	■ 050 / ☐ 052	00265	2024	- 002	- 01

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric – Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Turbine Trip and Automatic Scram due to Digital EHC Power Supply Intermittent Failure

CONDITION PRIOR TO EVENT

Unit: 2	Event Date: May 23, 2024	Event Time: 2223 CST
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 38 percent

No structures, systems, or components that were inoperable at the start of the event contributed to the event.

A. DESCRIPTION OF EVENT

At 2223 CST on May 23, 2024, with Quad Cities Unit 2 at 38 percent power, the reactor automatically scrammed due to a turbine [TRB] trip signal resulting in turbine stop valve closure, creating a valid reactor protection system signal. Reactor vessel level reached the low-level set-point following the scram, resulting in valid Group II and Group III containment actuation signals. The trip was not complex with all systems responding as expected post-trip. This report is being submitted per 10 CFR 50.73(a)(2)(iv)(A) for an event that resulted in manual or automatic actuation of the reactor protection system including a reactor scram, and containment isolation signals affecting more than one system.

Prior to the scram, the unit was reducing power for planned transformer maintenance. A turbine speed versus speed demand comparison generated a Speed Difference Trip and turbine trip signal due to an erroneous speed demand signal.

B. CAUSES OF EVENT

The cause of the event is an intermittent power supply failure that was not recognized as a single point vulnerability due to the diversity of the system. The turbine trip signal was generated by Digital Electro-Hydraulic Control (DEHC) [TG] turbine speed and speed demand mismatch logic. An intermittent failure of the channel <R1> core power supply [RJX] resulted in an erroneous speed demand value being generated. DEHC initiated a turbine trip due to a 100% actual speed compared to a 0% speed demand, which is larger than the 10% setpoint. While there was known degradation of this power supply, it was not recognized as a single point vulnerability to DEHC due to system redundancies.

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C. SAFETY ANALYSIS

SYSTEM DESIGN

Sections 7.2 and 10.4 of the Updated Final Safety Analysis Report (UFSAR) describe how the EHC system feeds into a turbine trip and reactor scram. "The electrohydraulic control (EHC) system compares generator stator current to the high pressure turbine exhaust (crossaround) pressure and operates these valves upon a mismatch indicative of a turbine generator load rejection (see Section 10.4). These pressure switches on each fast-acting solenoid provide signals to both RPS trip systems. The logic is a one-out-two-twice arrangement so that operation of any solenoid causes a single system trip, and the operation of one or more solenoids in each trip system initiates a scram."

SAFETY ANALYSIS

There were no safety consequences as a result of this event. The operators performed actions in accordance with procedures and training. An automatic scram occurred without complications due to a turbine trip caused by digital EHC channel <R1> core power supply intermittent failure. All expected Engineering Safety Feature (ESF) actuations occurred as designed to bring the reactor to a safe shutdown condition. The event was within the analysis of the UFSAR Chapter 15, UFSAR Chapter 6, and there were no radioactive releases. Unit 1 was not affected by the Unit 2 scram.

This is not a Safety System Functional Failure per NEI 99-02, Revision 7.

D. CORRECTIVE ACTIONS

Immediate:
1. DEHC Components were replaced: Power Supply and various input/output modules for channel <R1>.
2. Changed Speed Difference Trip setpoint from 10% to 110%.

Followup:
1. Change Speed Difference Trip setpoint for Unit 1 DEHC equivalent to the change implemented on Unit 2.
2. Increase frequency of preventative maintenance tasks for periodic replacement of <R1> power supplies.

E. PREVIOUS OCCURRENCES

The station events database, LERs, and INPO Industry Reporting Information System (IRIS) were reviewed for similar events at Quad Cities Nuclear Power Station in the last three years. A similar turbine trip and automatic reactor scram occurred on August 11, 2023 where a fault in the Essential Service System (ESS) [UJX] caused a feedwater heater (FWH) [HX] system transient. The event in this current Licensee Event Report was caused by a fault on the digital EHC system while pursuing a repair on a turbine control valve.

No other relevant events were identified during the previous 3-year history.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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			YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

F. COMPONENT FAILURE DATA

System: Electrohydraulic Control (EHC)
Component: GE Mark VI Rack Power Supply
Manufacturer: General Electric
Nomenclature: Power Supply
Model Number: IS2020RKPSG3A